

RVCalPy radial-velocity benchmark report

Broadening function: LL Aqr and AK For · Cross-correlation: 16 UVES high-velocity stars
RVCalPy, <https://github.com/atemelci/RVCalPy> · 2026 September 22

We compare RVCalPy velocities with published velocities of the same nights (LL Aqr, AK For) and of the same spectra (UVES sample). With the broadening function, the LL Aqr residuals scatter by 0.092 km s⁻¹ for the primary and 0.142 km s⁻¹ for the secondary ($\chi_r^2 = 1.03$ and 0.81). The AK For residuals scatter by 0.150 and 0.864 km s⁻¹ ($\chi_r^2 = 0.45$ and 2.54), with slopes of 0.9986 and 0.9923. With cross-correlation, the 16 UVES velocities differ from the published ones by a median $|\Delta|$ of 0.486 km s⁻¹ and a weighted offset of $+0.207 \pm 0.125$ km s⁻¹. Over the 919 km s⁻¹ velocity span the slope is 1.000171 ± 0.000479 .

1. What was tested

	LL Aqr	AK For	UVES sample
Method	Broadening function	Broadening function	Cross-correlation
Reference	Graczyk et al. (2016), Table 1	Helminiak et al. (2014), Table 2	Alazzawi et al. (2026), Table 3, V_r^{meas}
Instrument	HARPS	FEROS (14) + HARPS (2)	UVES, DIC1 390+580, red arm
Measured / compared	17 / 16 nights	16 / 14 nights	16 / 16 stars, one epoch each
Template	HIP 57757 (6093 K)	HIP 108870 (4676 K)	synthetic, 5215 K, log g 3.55
Window / step	500–550 nm, $\Delta v = 1$ km s ⁻¹	500–550 nm, $\Delta v = 2$ km s ⁻¹	500–550 nm
Period / velocity span	P = 20.1784 d	P = 3.98099 d	919 km s ⁻¹

Table 1. Material and settings. All spectra are public ESO archive phase-3 products; only epochs present in both data sets are compared.

2. How the numbers are defined

For each matched epoch we take the difference between our velocity and the published one, and add the two uncertainties in quadrature. Δ is the weighted mean offset, s the scatter, and χ_r^2 the reduced chi-square about Δ . The slope a and intercept b come from a weighted straight-line fit, and $a - 1$ is quoted as the amplitude error (BF) or the velocity-scale error (CCF):

$$d_i = v_i^{\text{ours}} - v_i^{\text{pub}}, \quad \sigma_i = \sqrt{\varepsilon_i^{2, \text{ours}} + \varepsilon_i^{2, \text{pub}}}, \quad w_i = \sigma_i^{-2}$$

$$\Delta = \frac{\sum w_i d_i}{\sum w_i}, \quad s = \sqrt{\frac{\sum (d_i - \bar{d})^2}{N - 1}}, \quad \chi_r^2 = \frac{1}{N - 1} \sum \left(\frac{d_i - \Delta}{\sigma_i} \right)^2, \quad v^{\text{ours}} = a v^{\text{pub}} + b$$

No χ_r^2 is given for the UVES comparison: both velocities are measured from the same exposures and no per-star error is published (the paper quotes a global ~ 0.5 km s⁻¹).

3. Broadening function: LL Aqr and AK For

	LL Aqr		AK For	
	Primary	Secondary	Primary	Secondary
Nights compared	16	16	14	14
Offset Δ (km s ⁻¹)	$+0.251 \pm 0.023$	$+0.294 \pm 0.040$	$+0.208 \pm 0.059$	$+0.318 \pm 0.130$
Scatter s (km s ⁻¹)	0.092	0.142	0.150	0.864
Reduced χ^2	1.03	0.81	0.45	2.54
Slope a	0.9997	0.9989	0.9986	0.9923
Amplitude error	-0.03%	-0.11%	-0.14%	-0.77%
Authors' own fit rms	0.053	0.049	0.082	0.096

Table 2. Broadening-function velocities against the published ones. The last row is the residual rms of the authors' own orbital fits.

LL Aqr (16 nights). For the primary and secondary respectively, the offsets are $+0.251 \pm 0.023$ and $+0.294 \pm 0.040$ km s⁻¹, the scatter is 0.092 and 0.142 km s⁻¹, and χ_r^2 is 1.03 and 0.81. The amplitude errors are -0.03% and -0.11%.

AK For (14 nights: 12 FEROS, 2 HARPS). For the primary and secondary respectively, the offsets are $+0.208 \pm 0.059$ and $+0.318 \pm 0.130$ km s⁻¹, the scatter is 0.150 and 0.864 km s⁻¹, and χ_r^2 is 0.45 and 2.54. The amplitude errors are -0.14% and -0.77%. The secondary residuals run from -1.5 km s⁻¹ near phase 0.1 to $+1.3$ km s⁻¹ beyond phase 0.6.

4. Cross-correlation: 16 UVES high-velocity stars

Star	RV, this work [km s ⁻¹]	V _r meas [km s ⁻¹]	V _r Gaia DR3 [km s ⁻¹]	Δ [km s ⁻¹]
TYC 7524-00065-1	+351.498 ± 0.027	+351.920	+350.953 ± 0.217	-0.422
TYC 7535-00160-1	+355.074 ± 0.037	+353.523	+355.069 ± 0.253	+1.551
C0213360-505024	+387.999 ± 0.027	+388.600	+388.222 ± 0.611	-0.601
C1012254-203007	+508.139 ± 0.035	+506.755	+508.138 ± 1.068	+1.384
C1032508-244851	+509.755 ± 0.029	+509.490	+515.354 ± 2.470	+0.265
C1126364-293415	+411.154 ± 0.019	+409.870	+409.674 ± 0.926	+1.284
C1250398-030748	+398.245 ± 0.031	+403.020	+396.221 ± 0.979	-4.775
C1302091-323721	+410.550 ± 0.021	+410.000	+409.817 ± 3.027	+0.550
C1304064-331913	+360.461 ± 0.019	+360.720	+357.825 ± 0.994	-0.259
C1330590-230955	+335.542 ± 0.027	+335.350	+334.919 ± 0.526	+0.192
TYC 7274-00734-1	+408.425 ± 0.020	+408.317	+407.955 ± 0.208	+0.108
C1519196-191359	-409.410 ± 0.026	-409.610	-409.694 ± 6.570	+0.200
C1536201-144228	-329.600 ± 0.023	-329.600	-328.647 ± 2.112	-0.000
C1919566-632839	+330.810 ± 0.032	+328.680	+330.399 ± 0.530	+2.130
C2136142-694908	+383.373 ± 0.017	+381.640	+382.882 ± 0.330	+1.733
TYC 8019-00159-1	+99.097 ± 0.035	+99.117	+98.450 ± 0.171	-0.020

Table 3. Rows follow the order of Table 3 in Alazzawi et al. (2026). Our velocities are barycentric; ± is the formal error of the Gaussian fit to the CCF peak. V_r^{meas} is their template-matching velocity from the same spectrum (no per-star error published), and $\Delta = \text{ours} - V_r^{\text{meas}}$. $V_r^{\text{Gaia DR3}}$ is listed for reference only and enters no statistic.

	All 16 stars	Excluding C1250398-030748
Weighted offset Δ (km s ⁻¹)	+0.207 ± 0.125	+0.539 ± 0.129
Median Δ (km s ⁻¹)	0.486	0.422
Scatter s (km s ⁻¹)	1.564	0.854
Within 1 km s ⁻¹	10 of 16	10 of 15
Velocity range (km s ⁻¹)	-409.4 to +509.8	-409.4 to +509.8
Slope a	1.000171 ± 0.000479	1.000771 ± 0.000482
Velocity-scale error a - 1	+0.017 ± 0.048%	+0.077 ± 0.048%

Table 4. Summary of Table 3, with and without the largest difference.

All 16 stars. The weighted offset is $+0.207 \pm 0.125$ km s⁻¹, the median |Δ| is 0.486 km s⁻¹, and the scatter is 1.564 km s⁻¹. 8 stars agree within 0.5 km s⁻¹ and 10 within 1 km s⁻¹. Across -409.410 to +509.755 km s⁻¹ (919 km s⁻¹), the slope is 1.000171 ± 0.000479 , a velocity-scale error of $+0.017 \pm 0.048\%$.

Largest difference. C1250398-030748 has Δ = -4.775 km s⁻¹; the next largest |Δ| is 2.130 km s⁻¹. Its spectrum has S/N 161 and a CCF peak significance of 31.8. Without this star the offset is $+0.539 \pm 0.129$ km s⁻¹, the scatter 0.854 km s⁻¹, and the slope 1.000771 ± 0.000482 .

References

- Alazzawi, J. M. K., Caffau, E., Bonifacio, P., et al. 2026, A&A, DOI 10.1051/0004-6361/202659789
Graczyk, D., Smolec, R., Gazeas, K., et al. 2016, A&A, 594, A92
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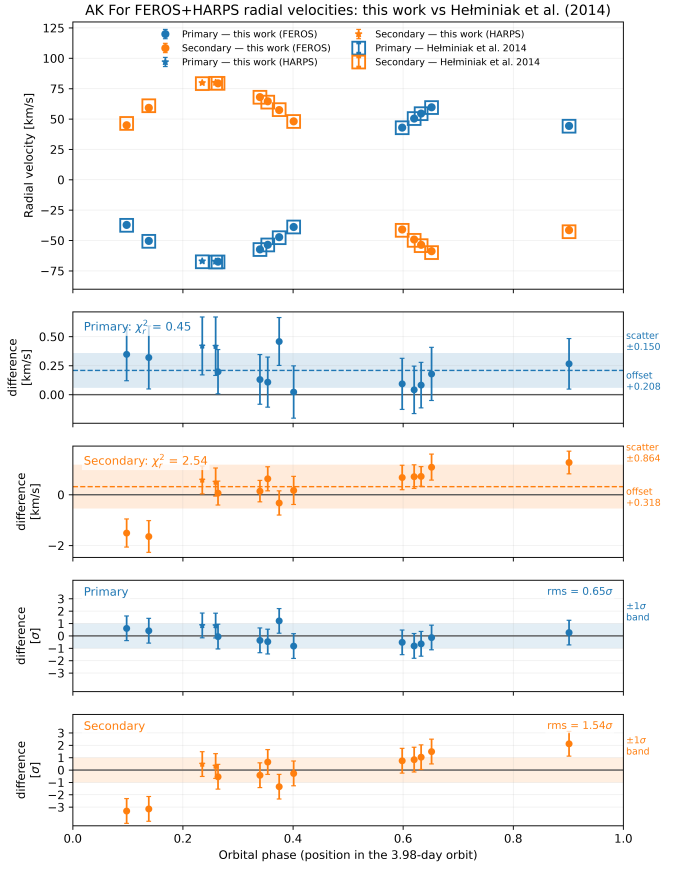
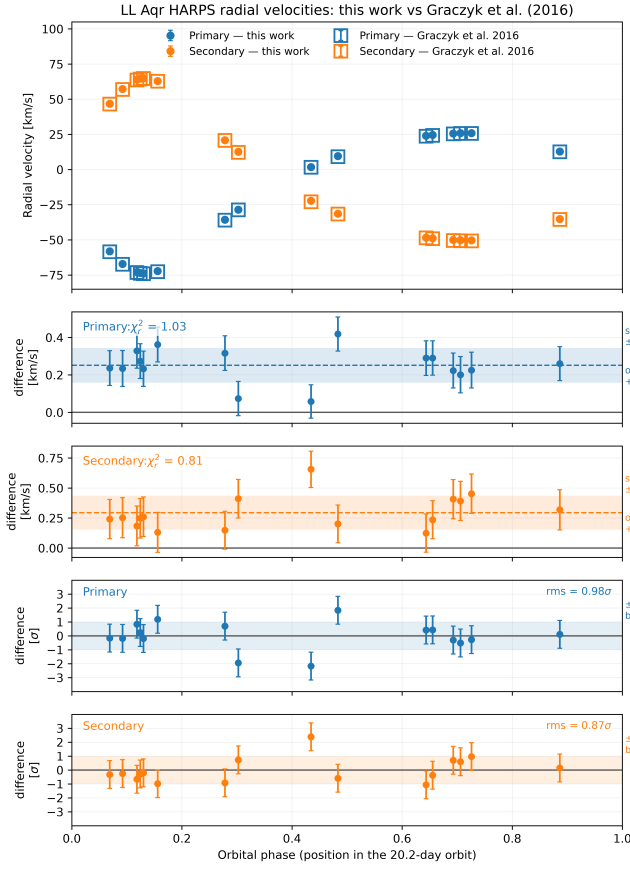


Figure 1. Broadening-function velocities against the published ones: LL Aqr versus Graczyk et al. 2016 (left, 16 nights) and AK For versus Helminiak et al. 2014 (right, 14 nights; circles FEROS, stars HARPS). Top: phased velocity curves (filled: this work, open: published). Middle: differences with the weighted offset (dashed) and scatter (band). Bottom: residuals in units of their own 1σ after removing the offset.